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Fluid end assembly

Abstract

A fluid end having its fluid flow bores sealed without threading a retaining nut into the walls of each bore. The fluid ends may be assembled using a plurality of different kits, each kit comprising a fluid end body, a component, a retainer element, and a fastening system. The retainer element holds the component within each of the bores formed in the fluid end body and the fastening system secures the retainer element to the body. The fastening system comprises a plurality of externally threaded studs, washers and nuts in some embodiments. In other embodiments, the fastening system comprises a plurality of screws.

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us-provisional-application US 63049974 20200709
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3005412	12/1960	Camp	N/A	N/A
3053500	12/1961	Atkinson	N/A	N/A
3146724	12/1963	Cornelson	N/A	N/A
3427988	12/1968	Redman et al.	N/A	N/A
3679332	12/1971	Yohpe	N/A	N/A
4432386	12/1983	Pacht	N/A	N/A
4520837	12/1984	Cole et al.	N/A	N/A
4758135	12/1987	Woodward et al.	N/A	N/A
4768933	12/1987	Stachowiak	N/A	N/A
4771801	12/1987	Crump et al.	N/A	N/A
4773833	12/1987	Wilkinson et al.	N/A	N/A
4861241	12/1988	Gamboa et al.	N/A	N/A
5073096	12/1990	King et al.	N/A	N/A
5226445	12/1992	Surjaatmadja	N/A	N/A

5362215	12/1993	King	N/A	N/A
6167959	12/2000	Bassinger et al.	N/A	N/A
7290560	12/2006	Orr et al.	N/A	N/A
7296591	12/2006	Moe et al.	N/A	N/A
7335002	12/2007	Vicars	N/A	N/A
7506574	12/2008	Jensen et al.	N/A	N/A
7866346	12/2010	Walters	N/A	N/A
8100407	12/2011	Stanton	277/522	F16J 15/183
8360094	12/2012	Steinbock et al.	N/A	N/A
8365754	12/2012	Riley et al.	N/A	N/A
9188121	12/2014	Dille	N/A	N/A
10221847	12/2018	Dyer	N/A	N/A
10760567	12/2019	Salih et al.	N/A	N/A
2004/0234404	12/2003	Vicars	N/A	N/A
2006/0002806	12/2005	Baxter	N/A	N/A
2009/0123303	12/2008	Ohnishi	N/A	N/A
2011/0189040	12/2010	Vicars	N/A	N/A
2011/0206547	12/2010	Kim	N/A	N/A
2011/0236238	12/2010	Cordes	N/A	N/A
2012/0141308	12/2011	Saini	N/A	N/A
2013/0202458	12/2012	Byrne	N/A	N/A
2013/0319220	12/2012	Luharuka	N/A	N/A
2014/0348677	12/2013	Moeller	N/A	N/A
2015/0132157	12/2014	Whaley	N/A	N/A
2017/0218951	12/2016	Graham	N/A	N/A
2018/0363642	12/2017	Salih et al.	N/A	N/A
2019/0011051	12/2018	Yeung	N/A	N/A
2020/0232455	12/2019	Blume	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO2014144113	12/2013	WO	N/A

OTHER PUBLICATIONS

GD Energy Products, LLC (Petitioner) v. Kerr Machine Co. (Patent Owner) Petition to Institute an Inter Partes Review for U.S. Pat. No. 12,152,582, Feb. 6, 2025, 165 pages. cited by applicant

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Background/Summary

SUMMARY

(1) The present disclosure is directed to a fluid end assembly. The fluid end assembly comprises a housing having an external surface and an internal chamber, a first conduit formed in the housing and having first and second sections, each section independently interconnecting the internal chamber and the external surface, and a second conduit formed in the housing, intersecting the first conduit and having third and fourth sections, each section independently interconnecting the

internal chamber and the external surface. The fluid end assembly also comprises a non-threaded component installed within the third section, and a plurality of packing seals installed within the component. The fluid end assembly further comprises a retainer engaged with the component, and a fastening system installed within the retainer and the housing and configured to releasably hold the retainer against the component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a partially exploded view of a first embodiment of a fluid end of the present invention. FIG. 1 shows a suction and discharge end of the fluid end.
- (2) FIG. 2 is a partially exploded view of a plunger end of the fluid end body shown in FIG. 1.
- (3) FIG. 3 is a cross-sectional view of the fluid end shown in FIG. 1, taken along line A-A.
- (4) FIG. 4 is a partially exploded view of a second embodiment of a fluid end.
- (5) FIG. 4 shows a suction and discharge end of the fluid end.
- (6) FIG. 5 is a partially exploded view of a plunger end of the fluid end body shown in FIG. 4.
- (7) FIG. 6 is a cross-sectional view of the fluid end shown in FIG. 4, taken along line B-B.
- (8) FIG. 7 is a partially exploded view of a third embodiment of a fluid end. FIG. 7 shows a suction and discharge end of the fluid end.
- (9) FIG. 8 is a partially exploded view of a plunger end of the fluid end body shown in FIG. 7.
- (10) FIG. 9 is a partially exploded view of a fourth embodiment of a fluid end. FIG. 9 shows a suction and discharge end of the fluid end.
- (11) FIG. 10 is a partially exploded view of a plunger end of the fluid end body shown in FIG. 9.
- (12) FIG. 11 is a cross-sectional view of the fluid end shown in FIG. 9, taken along line C-C.
- (13) FIG. 12 is a partially exploded view of a fifth embodiment of a fluid end. FIG. 12 shows a suction and discharge end of the fluid end.
- (14) FIG. 13 is a cross-sectional view of the fluid end shown in FIG. 12, taken along line D-D.
- (15) FIG. 14 is a partially exploded view of a sixth embodiment of a fluid end. FIG. 14 shows a suction and discharge end of the fluid end.
- (16) FIG. 15 is a side elevational view of one of the plurality of studs for use with the fluid ends.
- (17) FIG. 16 is a top plan view of a seventh embodiment of a fluid end.
- (18) FIG. 17 is a partially exploded front perspective view of the fluid end shown in FIG. 16.
- (19) FIG. 18 is a partially exploded rear perspective view of the fluid end shown in FIG. 16.
- (20) FIG. 19 is a cross-sectional view of the fluid end shown in FIG. 16, taken along line E-E.
- (21) FIG. 20 is a cross-sectional view of the fluid end shown in FIG. 16, taken along line F-F.
- (22) FIG. 21 is an enlarged view of area G shown in FIG. 19.
- (23) FIG. 22 is an enlarged view of area H shown in FIG. 19.
- (24) FIG. 23 is an enlarged view of area I shown in FIG. 20.
- (25) FIG. 24 is an enlarged view of area J shown in FIG. 20.
- (26) FIG. 25 is an enlarged view of area K shown in FIG. 21.
- (27) FIG. 26 is an enlarged view of area L shown in FIG. 22.
- (28) FIG. 27 is a top perspective view of a discharge plug shown installed within the fluid end in FIGS. 19 and 20.
- (29) FIG. 28 is a cross-sectional view of the discharge plug shown in FIG. 29, taken along line M-M.
- (30) FIG. 29 is a side elevational view of the discharge plug shown in FIG. 27.
- (31) FIG. 30 is a top perspective view of a suction plug shown installed within the fluid end in FIGS. 19 and 20.
- (32) FIG. 31 is a cross-sectional view of the suction plug shown in FIG. 32, taken along line N-N.

- (33) FIG. 32 is a side elevational view of the suction plug shown in FIG. 30.
- (34) FIG. 33 is a top perspective view of a retainer shown attached to the fluid end in FIGS. 19 and 20.
- (35) FIG. 34 is a top perspective view of a retainer nut shown attached to the fluid end in FIGS. 19 and 20.
- (36) FIG. 35 is a bottom perspective view of the retainer nut shown in FIG. 34.
- (37) FIG. 36 is a rear perspective view of a stuffing box shown installed within the fluid end in FIGS. 19 and 20.
- (38) FIG. 37 is a front perspective view of the stuffing box shown in FIG. 36.
- (39) FIG. 38 is a rear perspective view of a retainer shown attached to the fluid end in FIGS. 19 and 20.
- (40) FIG. 39 is a front perspective view of the retainer shown in FIG. 38.
- (41) FIG. 40 is a side perspective view of the fluid end shown in FIG. 16 attached to a power end.
- (42) FIG. 41 is a cross-sectional view of an eighth embodiment of a fluid end.
- (43) FIG. 42 is an enlarged cross-sectional view of the plunger bore of the fluid end shown in FIG. 41. The cross-section is taken along a line extending through a different plane than that shown in FIG. 41.
- (44) FIG. 43 is an enlarged cross-sectional view of the plunger bore of the fluid end shown in FIG. 41. The cross-section is taken along a line extending through a different plane than that shown in FIGS. 41 and 42.
- (45) FIG. 44 is a perspective view of a second surface of another embodiment of a stuffing box.
- (46) FIG. 45 is a perspective view of a first surface of the stuffing box shown in FIG. 44.
- (47) FIG. 46 is a side elevational view of the second surface of the stuffing box shown in FIG. 44.
- (48) FIG. 47 is a cross-sectional view of the stuffing box shown in FIG. 46, taken along line O-O.
- (49) FIG. 48 is a top plan view of another embodiment of a retainer.
- (50) FIG. 49 is a perspective view of a second surface of the retainer shown in FIG. 48.
- (51) FIG. 50 is a side elevational view of the second surface of the retainer shown in FIG. 48.
- (52) FIG. 51 is a cross-sectional view of the retainer shown in FIG. 50, taken along line P-P.
- (53) FIG. 52 is a perspective view of a first surface of the retainer shown in FIG. 48.
- (54) FIG. 53 is a side elevational view of the first surface of the retainer shown in FIG. 48.
- (55) FIG. 54 is a cross-sectional view of the retainer shown in FIG. 53, taken along line Q-Q.

DETAILED DESCRIPTION

- (56) Fluid end assemblies are typically used in oil and gas operations to deliver highly pressurized corrosive and/or abrasive fluids to piping leading to the wellbore. The assemblies are typically attached to power ends run by engines. The power ends reciprocate plungers within the assemblies to pump fluid throughout the fluid end. The power end used with the fluid end typically has a power output of at least 2,250 horsepower during hydraulic fracturing operations.
- (57) Fluid may be pumped through the fluid end at pressures that range from 5,000-15,000 pounds per square inch (psi). Fluid used in high pressure hydraulic fracturing operations is typically pumped through the fluid end at a minimum of 8,000 psi; however, fluid will normally be pumped through the fluid end at pressures around 10,000-15,000 psi during such operations.
- (58) In fluid end assemblies known in the art, the fluid flow passages or bores formed within the fluid end body are typically sealed by inserting a plug into each bore. A large retaining nut is then installed into each bore above the plug. The retaining nuts typically thread into internal threads formed in the walls of each bore.
- (59) In operation, the high level of fluid pressure pumping throughout the fluid end may cause the retaining nuts to back off or unthread from their installed position. When a retaining nut unthreads from its installed position, the plug it was retaining may be displaced by fluid pressure. Displacement of the plug allows fluid to leak around the plug and erode the walls of the bore. The internal threads formed in the bores for engagement with the retaining nuts are also known to crack

over time. Erosion of the bore walls or cracking of the internal threads typically requires repair or replacement of the fluid end.

(60) The present invention is directed to a plurality of different fluid ends having bores sealed without threading retaining nuts into the walls of each bore. As a result, the fluid ends of the present invention do not have internal threads formed in their bores proximate the bore openings. Removal of the internal threads eliminates the problems associated with the internal thread failures and the retaining nuts becoming unthreaded from the bores.

(61) With reference to FIGS. 1 and 3, a first embodiment of a fluid end **100** is shown. The fluid end **100** comprises a housing or fluid end body **102** having a flat external surface **104** and a plurality of first and second bores **106**, **108** formed adjacent one another therein, as shown in FIG. 1. The first bores **106** may also be referred to as vertical conduits, and the second bores **108** may also be referred to as horizontal conduits. Preferably, the number of first bores **106** equals the number of second bores **108**. More preferably, each first bore **106** intersects its paired second bore **108** within the fluid end body **102** to form an internal chamber **112**, as shown in FIG. 3. The first bore **106** comprises first and second sections positioned on opposite sides of the internal chamber **112**. Likewise, the second bore **108** comprises third and fourth sections positioned on opposite sides of the internal chamber **112**.

(62) FIG. 1 shows five first and second bores **106**, **108**. In alternative embodiments, the number of sets of paired first and second bores in the fluid end body may be greater than five, or less than five. Thus, FIG. 4 shows a fluid end body that includes three sets of paired first and second bores. Each bore of each set of paired bores **106** and **108** terminates in a corresponding opening **110** formed in the external surface **104**. The bores **106** and **108** and openings **110** exist in one-to-one relationship. A plurality of internally threaded openings **144** are formed in the body **102** and uniformly spaced around each bore opening **110**, as shown in FIG. 1.

(63) With reference to FIG. 3, each second bore **108** may have an intake opening **118** formed proximate the bottom end of the fluid end body **102**. Each intake opening **118** is connected in one-to-one relationship to a corresponding coupler or pipe. These couplers or pipes are fed from a single common piping system (not shown).

(64) A pair of valves **120** and **122** are positioned within each second bore **108**. The valves **120**, **122** route fluid flow within the body **102**. The intake valve **120** blocks fluid backflow through the intake opening **118**. The discharge valve **122** regulates fluid through one or more discharge openings **126**. A plurality of couplers **127** may be attached to each discharge opening **126** for connection to a piping system (not shown), as shown in FIG. 1.

(65) Continuing with FIGS. 1 and 3, the fluid end **100** further comprises a plurality of sets of components **128** and **130**. The number of sets preferably equals the number of sets of paired first and second bores **106** and **108** formed in the body **102**. The component **128** is positioned within a first bore **106**, and the component **130** is positioned within its paired second bore **108**. In one embodiment, the component **128** is a suction plug and the component **130** is a discharge plug. Each of the components **128** and **130** are substantially identical in shape and construction, and each is sized to fully block fluid flow within the respective bore **106**, **108**. A seal **136** is positioned around the outer surface of each component **128**, **130** to block fluid from leaking from the bores **106**, **108**.

(66) Each of the components **128** and **130** comprises a first section **138** joined to a second section **140**. The first section **138** has a footprint sized to cover the bore opening **110** and the second section **140** is configured for removable receipt within one of the bores **106**, **108**. In one embodiment, the first section **138** is an enlarged plate and the second section **140** is a plug sized to be closely received within one of the bores **106**, **108**. When the component **128** or **130** is installed within one of the bores **106**, **108**, the first section **138** engages with the external surface **104** of the body **102**. This engagement prevents longitudinal movement of the second section **140** within the bore **106** or **108** as shown in FIG. 3.

(67) With reference to FIG. 1, the first section **138** may be formed as a circular structure having a

plurality of notches **142** cut from its outer periphery. When each of the first sections **138** is engaged with the external surface **104** of the body **102**, each of the notches **142** partially surrounds one of the openings **144** spaced around each bore opening **110**.

(68) Continuing with FIGS. **1** and **3**, once each component **128**, **130** is installed in the fluid end body **102**, each of the components **128**, **130** is secured in place by a retainer element **132** in a one-to-one relationship. Each retainer element **132** has a footprint sized to fully cover the first section **138** of the components **128** and **130**. The retainer elements **132** shown in FIG. **1** are flat and cylindrical. A plurality of openings **146** are formed about the periphery of each retainer element **132**. Each opening **146** is alignable with a corresponding one of the openings **144** in a one-to-one relationship.

(69) Each of the retainer elements **132** is secured to the fluid end body **102** using a fastening system **134**. The fastening system comprises a plurality of studs **148**, a plurality of washers **150**, and a plurality of nuts **152**. Each stud **148** is externally threaded adjacent its first end **149**, while each opening **144** has internal threads that mate with those of the stud **148**. Each stud **148** may be threaded into place within a corresponding one of the openings **144**, in a one-to-one relationship.

(70) Once a first stud **148** has been installed in the body **102** at its first end **149**, its opposed second end **151** projects from the body's external surface **104**. When each component **128** is positioned within its bore **106**, each of its notches **142** at least partially surrounds a corresponding one of the studs **148**. Likewise, when each component **130** is positioned within its bore **108**, each of its notches **142** at least partially surrounds a corresponding one of the studs **148**.

(71) Each peripheral opening **146** formed in each of the retainer elements **132** is registerable with a corresponding one of the studs **148**. The plurality of washers **150** and nuts **152** may be installed and torqued on each one of the studs **148**. The plurality of washers **150** and nuts **152** hold the retainer element **132** against the first section **138** of the components **128**, **130** and hold the first section **138** against the external surface **104** of the fluid end body **102**. Because each of the retainer elements **132** is attached to the fluid end body **102** using the fastening system **134**, no external threads are formed on the outer surface of each retainer element **132**. Likewise, no internal threads are formed within the walls of each bore **106**, **108**.

(72) With reference to FIGS. **2** and **3**, a plunger end **154** of the fluid end **100** is shown. The plurality of first bores **106** terminate at openings **156** formed on the external surface **104** of the plunger end **154**. An internal seat **159** is formed in the walls of each of the bores **106** proximate each of the bore openings **156**. A plurality of threaded openings **161** are formed in each of the internal seats **159**, as shown in FIG. **2**.

(73) A component **158** is positioned within each first bore **106** through each of the openings **156**. Each of the components **158** is tubular and sized to be closely received within each bore **106**. In one embodiment, the components **158** are stuffing box sleeves.

(74) With reference to FIG. **3**, each of the components **158** may have a first or rear section **160** that joins a second or front section **162** via a tapered section **164**. The first section **160** may have a larger diameter than the second section **162**. When each of the components **158** is installed within each of the bores **106**, the tapered section **164** engages a tapered seat **166** formed in the walls of each bore **106**. This engagement prevents longitudinal movement of each component **158** within each bore **106**. A seal **167** is positioned around the outer surface of the second section **162** of each of the components **158** in order to block fluid from leaking from the bores **106**.

(75) Once installed within the body **102**, each component **158** is secured in place by a retainer element **170** in a one-to-one relationship. Each of the retainer elements **170** is sized to be closely received within each bore **106** and engage a top surface **171** of each component **158**, as shown in FIG. **3**. Each of the retainer elements **170** shown in FIG. **2** has a cylindrical body and a threaded central opening **172**. A plurality of openings **174** are formed about the periphery of each of the retainer elements **170**. The openings **174** are uniformly spaced around each central opening **172**.

(76) A plurality of ports **175** may be formed in an outer surface of each retainer element **170** that

are orthogonal to the plurality of openings **174**. At least one seal **176** may also be disposed around the outer surface of each of the retainer elements **170**. The seal **176** helps block fluid from leaking from the bores **106**.

(77) Each of the retainer elements **170** is secured to the fluid end body **102** using a fastening system **178**. The fastening system **178** comprises a plurality of threaded screws **180**. The screws **180** are preferably socket-headed cap screws.

(78) The fastening system **178** secures each retainer element **170** to each internal seat **159**. When each retainer element **170** is positioned within each bore **106**, each of the peripheral openings **174** is alignable with a corresponding one of the openings **161** in a one-to-one relationship. Each of the screws **180** is registerable within one of the openings **161** in the seat **159** and one of the peripheral openings **174** in the retainer element **170**.

(79) The screws **180** may be torqued as desired to tightly attach each of the retainer elements **170** to each internal seat **159** and securely hold each component **158** within each bore **106**. Because each of the retainer elements **170** is attached to the fluid end body **102** using the fastening system **178**, no external threads are formed on the outer surface of each of the retainer elements **170**. Likewise, no internal threads are formed within the walls of each bore **106** on the plunger end **154** of the body **102**.

(80) Continuing with FIGS. 2-3, a plurality of packing seals **181** may be positioned within each of the components **158** and each of the retainer elements **170** to prevent fluid from leaking from the bores **106**. At least one of the packing seals **181** may have a plurality of ports **179** formed in its outer periphery, as shown in FIG. 2. The ports **179** provide an exit for fluid trapped within the packing seals **181**. Fluid exiting the ports **179** may exit the retainer element **170** through the ports **175**.

(81) A packing nut **182** may also be threaded into the central opening **172** of each of the retainer elements **170** in a one-to-one relationship. The packing nut **182** has a threaded section **183** joined to a body **184**. The body **184** shown in FIG. 2 is cylindrical. However, the body **184** may also be square or rectangular shaped. A central passage **185** extends through the threaded section **183** and the body **184**. The threaded section **183** of the packing nut **182** is threaded into the central opening **172** of the retainer element **170**.

(82) When installed within each of the retainer elements **170**, each of the packing nuts **182** engages with and compresses the packing seals **181** installed within each component **158** and retainer element **170**, as shown in FIG. 3. Compression of the packing seals **181** helps prevent fluid from leaking past the seals **181**. A seal **186** may also be positioned within the central passage **185** of each of the packing nuts **182** to further seal fluid from leaking from the bores **106**.

(83) A plurality of holes **187** are formed around the outer surface of each of the packing nut bodies **184**. The holes **187** serve as connection points for a spanner wrench that may be used to tightly thread the packing nut **182** into the central opening **172** of each of the retainer elements **170**.

(84) A plunger **188** may also be installed within each bore **106** in a one-to-one relationship. When a plunger **188** is installed within a bore **106**, the plunger **188** is positioned within the component **158**, the retainer element **170**, and the packing nut **182**, as shown in FIG. 3. Each of the plungers **188** projects from the plunger end **154** of the fluid end body **102** and is attached to a separate power end. As discussed above, the power end reciprocates each of the plungers **188** within the fluid end body **102** so as to pump fluid throughout the body. Each of the plungers **188** may be attached to the power end via a clamp **190** in a one-to-one relationship.

(85) Several kits are useful for assembling the fluid end **100**. A first kit comprises a plurality of the components **128** or **130**, a plurality of the retainer elements **132**, and the fastening system **134**. A second kit may comprise the plurality of components **158**, a plurality of the retainer elements **170**, and the fastening system **178**. The second kit may further comprise a plurality of the packing seals **181**, a plurality of the packing nuts **182**, and a plurality of the plungers **188**. Each of the kits may be assembled using the fluid end body **102**.

(86) With reference to FIGS. 4 and 6, a second embodiment of a fluid end **200** is shown. The fluid end **200** comprises a housing or fluid end body **202** having a flat external surface **204** and a plurality of first and second bores **206**, **208** formed adjacent one another therein, as shown in FIG. 4. Each bore of each set of paired bores **206** and **208** terminates in a corresponding opening **210** formed in the external surface **204**. A plurality of threaded openings **211** are formed in the body **202** and uniformly spaced around each opening **210**. The internal functions of the fluid end **200** are identical to those described with reference to fluid end **100**, shown in FIG. 3.

(87) The fluid end **200** further comprises a plurality of sets of components **212** and **214**. The number of sets preferably equals the number of set of paired first and second bores **206** and **208** formed in the body **202**. The component **212** is positioned within a first bore **206**, and the component **214** is positioned within its paired second bore **208**. In one embodiment, the component **212** is a suction plug and the component **214** is a discharge plug.

(88) Each of the components **212** and **214** is substantially identical in shape and construction, and is sized to fully block fluid flow within the respective bore **206**, **208**. A seal **216** is positioned around the outer surface of each component **212**, **214** to block fluid from leaking from the bores **206**, **208**.

(89) As shown in FIG. 4, a top surface **213** of each component **212**, **214** may sit flush with the external surface **204** of the body **202** when installed within a respective bore **206**, **208**. Each of the components **212** and **214** may engage with internal seats (not shown) formed in the walls of each of the bores **206**, **208**. Such engagement helps prevent longitudinal movement of the components **212**, **214** within the respective bore **206**, **208**.

(90) Once installed within the fluid end body **202**, each component **212** and **214** is secured in place by a retainer element **218** in a one-to-one relationship. Each of the retainer elements **218** has a footprint sized to cover a single bore opening **210**. The retainer elements **218** shown in FIG. 4 are flat and cylindrical. A plurality of openings **220** are formed about the periphery of each retainer element **218**. Each peripheral opening **220** is alignable with a corresponding one of the openings **211** in a one-to-one relationship, as shown in FIG. 4.

(91) The retainer elements **218** are secured to the external surface **204** of the fluid end body **202** by a fastening system **222**. The fastening system **222** comprises a plurality of externally threaded studs **224**, a plurality of washers **226**, and a plurality of internally threaded nuts **228**. Each stud **224** is externally threaded adjacent its first end **230**, while each opening **211** has internal threads that mate with those of the stud **224**. Each stud **224** may be threaded into place within a corresponding one of the openings **211**, in a one-to-one relationship.

(92) Once a first stud **224** has been installed in the body **202** at its first end **230**, its opposed second end **232** projects from the body's external surface **204**. Each peripheral opening **220** formed in the retainer elements **218** is registerable with a corresponding one of the studs **224**. The plurality of washers **226** and nuts **228** may be installed and torqued on each of the studs **224**. The plurality of washers **226** and nuts **228** hold the retainer elements **218** against the external surface **204** of the fluid end body **202**. Because each of the retainer elements **218** is attached to the fluid end body **202** using the fastening system **222**, no external threads are formed on the outer surface of each retainer element **218**. Likewise, no internal threads are formed within the walls of each bore **206** and **208**.

(93) With reference to FIGS. 5-6, a plunger end **234** of the fluid end **200** is shown. The plurality of first bores **206** terminate at openings **236** formed on the external surface **204** of the plunger end **234**. The plunger end **234** of the fluid end body **202** is similar to the plunger end **154** of fluid end body **102**, shown in FIGS. 2-3, except that an internal seat **159** is not formed within each bore **206**. Instead, a plurality of internally threaded openings **238** are formed in the external surface **204** of the fluid end body **202** that are uniformly spaced around each bore opening **236**.

(94) A component **240** is positioned within each first bore **206** through each of the openings **236** in a one-to-one relationship. Each of the components **240** is tubular and sized to be closely received within each bore **206**. In one embodiment, the components **240** are stuffing box sleeves.

(95) With reference to FIG. 6, each of the components **240** may have a first or rear section **242** that joins a second or front section **244** via a tapered section **246**. The first section **242** may have a larger diameter than the second section **244**. When each of the components **240** is installed within each of the bores **206**, the tapered section **246** engages a tapered seat **248** formed in the walls of each bore **206**. This engagement prevents longitudinal movement of each component **240** within each bore **206**. A seal **250** is positioned around the outer surface of the second section **244** of each of the components **240** to block fluid from leaking from the bores **206**.

(96) Once installed within the body **202**, a top surface **252** of each of the components **240** may sit flush with the external surface **204** of the body **202**. Each of the components **240** is secured in place within each bore **206** by a retainer element **254** in a one-to-one relationship. The retainer elements **254** shown in FIG. 5 have a cylindrical body and a threaded central opening **256**. A plurality of openings **258** are formed about the periphery of each of the retainer elements **254**. The openings **258** are uniformly spaced around each central opening **256**.

(97) The retainer elements **254** are secured to the external surface **204** of the fluid end body **202** using a fastening system **260**. The fastening system **260** comprises a plurality of threaded screws **262**. The screws **262** are preferably socket-headed cap screws. When each retainer element **254** is positioned over each bore opening **236**, each of the peripheral openings **258** is alignable with a corresponding one of the openings **238** in a one-to-one relationship. Each of the screws **262** is registerable within one of the openings **238** in the body **202** and one of the peripheral openings **258** in each of the retainer elements **254**.

(98) The screws **262** may be torqued as desired to tightly attach each of the retainer elements **254** to the body **202** and securely hold each of the components **240** within each bore **206**. Because each of the retainer elements **254** is attached to the fluid end body **202** using the fastening system **260**, no external threads are formed on the outer surface of each retainer element **254**. Likewise, no internal threads are formed within the walls of each bore **206** on the plunger end **234** of the body **202**.

(99) Similar to the plunger end **154** shown in FIG. 2, a plurality of packing seals **264** may be positioned within each of the components **240**. A packing nut **266** may thread into the central opening **256** of each retainer element **254** and compress the packing seals **264**. A seal **267** may also be positioned within each packing nut **266**. Additionally, a plurality of plungers **268** may be disposed within each component **240**, retainer element **254**, and packing nut **266**. Each of the plungers **268** may be attached to a power end via a clamp **270**.

(100) In alternative embodiments, the components **212**, **214**, and **240** may not be flush with the external surface **204** of the body **202** when installed in the respective bores **206**, **208**. In such case, a flange or ledge may be formed on each of the retainer elements **218** or **254** on its side facing the component **212**, **214**, or **240**. The flange or ledge may be installed within the bores **206**, **208** so that it tightly engages the top surface **213** or **252** of the components **212**, **214**, or **240**.

(101) Likewise, if the components **212**, **214**, or **240** project from the external surface **204** of the body **202** when installed within the respective bores **206**, **208**, the retainer elements **218** or **254** can be modified to accommodate the component **212**, **214**, or **240**. For example, a cut-out may be formed in the retainer element **218** or **254** for closely receiving the portion of the component **212**, **214**, or **240** projecting from the body **202**. The area of the retainer element **218** or **254** surrounding the cut-out will engage the external surface **204** of the body **202**.

(102) Several kits are useful for assembling the fluid end **200**. A first kit comprises a plurality of the components **212** or **214**, a plurality of retainer elements **218**, and the fastening system **222**. A second kit may comprise the plurality of components **240**, a plurality of the retainer elements **254**, and the fastening system **260**. The second kit may further comprise a plurality of packing seals **264**, a plurality of packing nuts **266**, and a plurality of plungers **268**. Each of the kits may be assembled using the fluid end body **202**.

(103) Turning now to FIG. 7, a third embodiment of a fluid end **300** is shown. The fluid end **300** comprises a housing or fluid end body **302** having a flat external surface **304** and a plurality of first

and second bores **306**, **308** formed adjacent one another therein. Each bore of each set of paired bores **306** and **308** terminates in a corresponding opening **310** formed in the external surface **304**. A plurality of threaded openings **311** are formed in the body **302** and uniformly spaced around each bore opening **310**. The internal functions of the fluid end **300** are identical to those described with reference to fluid end **100**, shown in FIG. 3.

(104) The fluid end **300** further comprises a plurality of sets of components **312** and **314**. The number of sets preferably equals the number of set of paired first and second bores **306** and **308** formed in the body **302**. The component **312** is positioned within a first bore **306**, and the component **314** is positioned within its paired second bore **308**. In one embodiment, the component **312** is a suction plug and the component **314** is a discharge plug. A seal **315** is positioned around each of the components **312**, **314** to block fluid from leaking from the respective bores **306**, **308**.

(105) The components **312** and **314** have the same shape and construction as the components **212** and **214** shown in FIGS. 4 and 6. Each of the components **312** and **314** may engage with internal seats (not shown) formed in the walls of each of the bores **306**, **308**. Such engagement helps prevent longitudinal movement of the components **312**, **314** within the respective bores **306**, **308**.

(106) Once installed within the body **302**, a top surface **313** of each of the components **312**, **314** may sit flush with the external surface **304** of the body **302**. Each of the components **312**, **314** is secured within each respective bore **306**, **308** by a retainer element **316**. Each of the retainer elements **316** shown in FIG. 7 is a large rectangular plate having a footprint sized to cover a plurality of adjacent bore openings **310** at one time. A plurality of openings **318** are formed in each retainer element **316** that are alignable with a corresponding one of the openings **311** in a one-to-one relationship.

(107) Each of the retainer elements **316** is secured to the external surface **304** of the fluid end body **302** by a fastening system **320**. The fastening system **320** comprises a plurality of externally threaded studs **322**, a plurality of washers **324**, and a plurality of internally threaded nuts **326**. The fastening system **320** secures each of the retainer elements **316** on the fluid end body **302** in the same way as described with reference to the fastening system **222** used with the fluid end **200**.

(108) Because each of the retainer elements **316** is attached to the fluid end body **302** using the fastening system **320**, no external threads are formed in the retainer element **316**. Likewise, no internal threads are formed within the walls of each bore **306** and **308**. When the retainer elements **316** are installed on the fluid end body **302**, the edges of the retainer element **316** may extend far enough so as to sit flush with the edges of the fluid end body **302**. In alternative embodiments, the retainer element **316** may have different shapes or sizes. For example, the retainer element **316** may be large enough so as to cover an entire side surface of the fluid end body **302**. Alternatively, the retainer elements **316** may have rounded edges, as shown in FIG. 8.

(109) Turning to FIG. 8, a plunger end **330** of the fluid end **300** is shown. The plurality of first bores **306** terminate at openings **332** formed on the external surface **304** of the plunger end **330**. A plurality of internally threaded openings **334** are formed in the external surface **304** that are uniformly spaced around each bore opening **332**.

(110) A component **336** is positioned within each first bore **306** through each of the openings **332**. Each of the components **336** is tubular and sized to be closely received within each bore **306**. In one embodiment, the components **336** are stuffing box sleeves. The components **336** have the same shape and construction as the components **240**, shown in FIGS. 5-6.

(111) Once installed within the body **302**, a top surface **346** of each of the components **336** may sit flush with the external surface **304** of the body **302**. Each of the components **336** is secured within each bore **306** by a single retainer element **348**. The retainer element **348** shown in FIG. 8 is a large oval plate having a footprint sized to cover a plurality of adjacent bore openings **332** formed on the plunger end **330** of the fluid end body **302**. A plurality of openings **350** are formed in the retainer element **348** that are alignable with a corresponding one of the openings **334** in a one-to-one relationship.

(112) In alternative embodiments, the retainer element **348** may have different shapes or sizes. For example, the retainer element **348** may be large enough so as to cover an entire side surface of the fluid end body **302**. Alternatively, the retainer element **348** may have squared edges, as shown in FIG. 7.

(113) The retainer element **348** is secured to the external surface **304** of the fluid end body **302** by a fastening system **352**. The fastening system **352** comprises a plurality of screws **354**. The fastening system **352** secures the retainer element **348** on the fluid end body **302** in the same way as described with reference to the fastening system **260** used with the fluid end **200** and shown in FIGS. 5-6.

(114) Because the retainer element **348** is attached to the fluid end body **302** using the fastening system **352**, no external threads are formed in the retainer element **348**. Likewise, no internal threads are formed within the walls of each bore **306**.

(115) A central threaded opening **356** is formed in the center of each grouping of openings **350** in the retainer element **348**. The openings **356** are alignable with each bore opening **332** in a one-to-one relationship. A single packing nut **358** may thread into each central opening **356**. A seal **359** may be positioned within each packing nut **358**.

(116) Similar to the plunger end **234** shown in FIGS. 5-6, a plurality of packing seals **360** may be positioned within each component **336**. Each of the packing nuts **358** may compress the packing seals **360** when installed within the retainer element **348**. A plurality of plungers **362** may be disposed within each component **336**, the retainer element **348**, and each packing nut **358**. Each of the plungers **362** may be connected to a power end via a clamp **364**. A cross-sectional view of the fluid end **300** looks identical to the cross-sectional view of the fluid end **200**, shown in FIG. 6.

(117) Several kits are useful for assembling the fluid end **300**. A first kit comprises a plurality of the components **312** or **314**, a retainer element **316**, and the fastening system **320**. A second kit may comprise a plurality of the components **336**, a retainer element **348**, and the fastening system **352**. The second kit may further comprise a plurality of the packing seals **360**, a plurality of the packing nuts **358**, and a plurality of the plungers **362**. Each of the kits may be assembled using the fluid end body **302**.

(118) With reference to FIGS. 9 and 11, a fourth embodiment of a fluid end **400** is shown. The fluid end **400** comprises a housing or fluid end body **402** having a flat external surface **404** and a plurality of first and second bores **406**, **408** formed adjacent one another therein, as shown in FIG. 9. Each bore of each set of paired bores **406** and **408** terminates in a corresponding opening **410** formed in the external surface **404**. A plurality of threaded openings **411** are formed in the body **402** and uniformly spaced around each opening **410**. The internal functions of the fluid end **400** are identical to those described with reference to fluid end **100**, shown in FIG. 3.

(119) The fluid end **400** further comprises a plurality of sets of components **412** and **414**. The number of sets preferably equals the number of set of paired first and second bores **406** and **408** formed in the body **402**. The component **412** is positioned within a first bore **406**, and the component **414** is positioned within its paired second bore **408**. In one embodiment, the component **412** is a suction plug and the component **414** is a discharge plug. A seal **415** is positioned around the outer surface of each of the components **412**, **414** to block fluid from leaking from the respective bores **406**, **408**.

(120) The components **412** and **414** have substantially the same shape and construction as the components **212** and **214** shown in FIGS. 4 and 6. However, in contrast to the components **212**, **214**, each of the components **412** and **414** is joined to a single retainer element **416**.

(121) The components **412**, **414** may be welded or fastened to the center of the back surface of each retainer element **416**. Alternatively, each of the components **412** or **414** and a corresponding retainer element **416** may be machined as a single piece, as shown in FIG. 11. Each of the retainer elements **416** secures each of the components **412**, **414** within the respective bores **406**, **408**. The retainer elements **416** also prevent the components **412**, **414** from moving longitudinally within the

respective bores **406**, **408**.

(122) A plurality of openings **418** are formed about the periphery of each retainer element **416**. Each peripheral opening **418** is alignable with a corresponding one of the openings **411** in a one-to-one relationship, as shown in FIG. **9**.

(123) The retainer elements **416** are secured to the external surface **404** of the body **402** using a fastening system **420**. The fastening system **420** comprises a plurality of externally threaded studs **422**, a plurality of washers **424**, and a plurality of internally threaded nuts **426**. The fastening system **420** secures the retainer elements **416** to the fluid end body **402** in the same way as described with reference to the fastening system **222** used with the fluid end **200**.

(124) Because the retainer elements **416** are attached to the fluid end body **402** using the fastening system **420**, no external threads are formed in the retainer elements **416**. Likewise, no internal threads are formed within the walls of each bore **406** and **408**.

(125) Turning now to FIGS. **10-11**, a plunger end **430** of the fluid end **400** is shown. The plurality of first bores **406** terminate at openings **432** formed on the external surface **404** of the plunger end **430**. A plurality of internally threaded openings **434** are formed in the external surface **404** that are uniformly spaced around each bore opening **432**. A component **436** is positioned within each first bore **406** through each of the openings **432**. Each of the components **436** is tubular and sized to be closely received within each bore **406**. In one embodiment, the components **436** are stuffing box sleeves. The components **436** have substantially the same shape and construction as the components **240**, shown in FIGS. **5-6**. However, in contrast to the components **240**, each of the components **436** is joined to a single retainer element **438**.

(126) The components **436** may be welded or fastened to the center of the back surface of each retainer element **438**. Alternatively, each of the components **436** and a corresponding retainer element **438** may be machined as a single piece, as shown in FIG. **11**. Each of the retainer elements **438** secures each of the components **436** within the bores **406**. The retainer elements **438** also prevent the components **436** from moving longitudinally within the bores **406**.

(127) A threaded central opening **440** is formed within each retainer element **438**. A plurality of threaded openings **442** are formed about the periphery of each of the retainer elements **438** and are uniformly spaced around each central opening **440**. Each peripheral opening **442** is alignable with a corresponding one of the openings **434** in a one-to-one relationship, as shown in FIG. **10**.

(128) The retainer elements **438** are secured to the external surface **404** of the body **402** using a fastening system **444**. The fastening system **444** comprises a plurality of screws **446**. The fastening system **444** secures the retainer elements **438** to the fluid end body **402** in the same way as described with reference to the fastening system **260** used with the fluid end **200** and shown in FIGS. **5-6**.

(129) Because the retainer elements **438** are attached to the fluid end body **402** using the fastening system **444**, no external threads are formed in the retainer elements **416**. Likewise, no internal threads are formed within the walls of each bore **406** on the plunger end **430** of the body **402**.

(130) Like the plunger end **330** of fluid end **300**, the fluid end **400** may also comprise a plurality of packing seals **448**, a plurality of packing nuts **450**, each housing a seal **454**, and a plurality of plungers **456**. Each plunger **456** may be connected to a power end via a clamp **458**.

(131) Several kits are useful for assembling the fluid end **400**. A first kit comprises a plurality of the components **412** or **414**, a plurality of the retainer elements **416**, and the fastening system **420**. A second kit may comprise a plurality of the components **436**, a plurality of the retainer elements **438**, and the fastening system **444**. The second kit may further comprise a plurality of the packing seals **448**, a plurality of the packing nuts **450** and a plurality of the plungers **456**. Each of the kits may be assembled using the fluid end body **402**.

(132) With reference to FIGS. **12-13**, a fifth embodiment of a fluid end **500** is shown. The fluid end **500** comprises a housing or fluid end body **502** having a flat external surface **504** and a plurality of first and second bores **506**, **508** formed adjacent one another therein, as shown in FIG. **12**. Each

bore of each set of paired bores **506** and **508** terminates in a corresponding opening **510** formed in the external surface **504**. A plurality of threaded openings **511** are formed in the body **502** and uniformly spaced around each opening **510**. The internal functions of the fluid end **500** are identical to those described with reference to fluid end **100**, shown in FIG. 3.

(133) The fluid end **500** further comprises a plurality of sets of components **512** and **514**. The number of sets preferably equals the number of set of paired first and second bores **506** and **508** formed in the body **502**. The component **512** is positioned within a first bore **506**, and the component **514** is positioned within its paired second bore **508**. In one embodiment, the component **512** is a suction plug and the component **514** is a discharge plug. The components **512** and **514** have the same shape and construction as the components **212** and **214** shown in FIGS. 4 and 6. A seal **516** is positioned around the outer surface of each component **512**, **514** to block fluid from leaking from the bores **506**, **508**.

(134) As shown in FIG. 12, a top surface **513** of each of the components **512**, **514** may sit flush with the external surface **504** of the body **502** when installed within a respective bore **506**, **508**. Each of the components **512** and **514** may engage with internal seats (not shown) formed in the walls of each of the bores **506**, **508**. Such engagement helps prevent longitudinal movement of the components **512**, **514** within the respective bore **506**, **508**.

(135) Once installed within the fluid end body **502**, each component **512** and **514** is secured in place by a retainer element **518** in a one-to-one relationship. Each of the retainer elements **518** has a footprint sized to cover a single bore opening **510**. The retainer elements **518** shown in FIG. 12 are flat and cylindrical and each have a central threaded opening **519**. A plurality of openings **520** are formed about the periphery of each retainer element **518** and are uniformly spaced around each central opening **519**. Each peripheral opening **520** is alignable with a corresponding one of the openings **511** in a one-to-one relationship, as shown in FIG. 12.

(136) The retainer elements **518** are secured to the external surface **504** of the fluid end body **502** by a fastening system **522**. The fastening system **522** comprises a plurality of externally threaded studs **524**, a plurality of washers **526**, and a plurality of internally threaded nuts **528**. The fastening system **522** secures the retainer elements **518** to the fluid end body **502** in the same way as described with reference to the fastening system **222** used with the fluid end **200** shown in FIGS. 4 and 6.

(137) Each central opening **519** formed in each retainer element **518** is alignable with each corresponding bore opening **510** in a one-to-one relationship. A retaining nut **530** may thread into each central opening **519** to cover each bore opening **510**. Using a threaded retaining nut **530** with the retainer element **518** allows access to each bore opening **510** without having to remove the retainer elements **518** from the fluid end body **502**.

(138) While the fluid end **500** uses a threaded retaining nut **530**, the retaining nut **530** is not threaded into the walls of the bores **506**, **508**. Thus, any failures associated with the retaining nut **530** may be experienced in the retainer element **518**, which is easily replaceable. This similar configuration is used on the plunger end **234** of the fluid end **200** shown in FIGS. 5-6. Such configuration is shown again on a plunger end **532** of the fluid end body **502** in FIG. 13.

(139) A kit is useful for assembling the fluid end **500**. The kit may comprise a plurality of the components **512** or **514**, a plurality of the retainer elements **518**, and the fastening system **522**. The kit may further comprise a plurality of retaining nuts **530**. The kit may be assembled using the fluid end body **502**.

(140) Turning now to FIG. 14, a sixth embodiment of a fluid end **600** is shown. The fluid end **600** comprises a housing or fluid end body **602** having a flat external surface **604** and a plurality of first bores (not shown) and second bores **608** formed adjacent one another therein. Each bore of each set of paired bores terminates in a corresponding opening **610** formed in the external surface **604**. A plurality of threaded openings **611** are formed in the body **602** and uniformly spaced around each opening **610**. The internal functions of the fluid end **600** are identical to those described with

reference to fluid end **100**, shown in FIG. 3.

(141) The fluid end **600** further comprises a plurality of sets of components **614**. The component **614** is positioned within a second bore **608**. The components positioned within each first bore are not shown in FIG. 14. However, such components are identical in shape and construction to the components **614**.

(142) The number of sets of components preferably equals the number of set of paired first bores (not shown) and second bores **608** formed in the body **602**. In one embodiment, the component positioned within a first bore is a suction plug, and the component **614** positioned within its paired second bore **608** is a discharge plug. The components **614** have a substantially similar shape and construction as the components **212** and **214** shown in FIGS. 4 and 6, except that a threaded hole **616** is formed in a top surface **613** of each component **614**. A seal **618** is positioned around the outer surface of each component **614** to block fluid from leaking from the bores **608**.

(143) The top surface **613** of each component **614** may sit flush with the external surface **604** of the body **602** when installed within a bore **608**. Each of the components **614** may engage with internal seats (not shown) formed in the walls of each of the bores **608**. This engagement helps prevent longitudinal movement of the components **614** within the bore **608**. Likewise, the components positioned within the first bores (not shown) may engage internal seats formed within the walls of the first bores.

(144) Once installed within the fluid end body **602**, each component **614** is secured by a retainer element **620** in a one-to-one relationship. Likewise, the components positioned within the first bores (not shown) are each secured by one of the retainer elements **620**. Each of the retainer elements **620** has a footprint sized to cover a single bore opening **610**. The retainer elements **620** shown in FIG. 14 are flat and cylindrical and each have a central threaded opening **622**. A plurality of openings **624** are formed about the periphery of each retainer element **620** and are uniformly spaced around each central opening **622**. Each peripheral opening **624** is alignable with a corresponding one of the openings **611** in a one-to-one relationship.

(145) The retainer elements **620** are secured to the external surface **604** of the fluid end body **602** by a fastening system **626**. The fastening system **626** comprises a plurality of externally threaded studs **628**, a plurality of washers (not shown), and a plurality of internally threaded nuts **630**. The fastening system **626** secures the retainer elements **620** to the fluid end body **602** in the same way as described with reference to the fastening system **222** used with the fluid end **200** shown in FIGS. 4 and 6.

(146) The fastening system **626** may further comprise a plurality of eye bolts **632**, a plurality of handles **634**, and a cable **636**. Each eye bolt **632** has external threads **638** formed on its first end and an eye **640** formed on its opposite second end. The threaded end **638** of each eye bolt **632** threads into each hole **616** formed in each component **614** in a one-to-one relationship. Once installed within each hole **614**, the eye **640** of each eyebolt **632** projects through the central opening **622** formed in each retainer element **620**.

(147) Each of the handles **634** has a threaded section **642** joined to a cylindrical body **644**. A central passage **646** extends through the threaded section **642** and the body **644**. Each of the threaded sections **642** may be installed within the central opening **622** of each of the retainer elements **620** such that each eye bolt **632** is disposed within the central passage **646**. Once one of the handles **634** is installed in a retainer element **620**, the eye bolt **632** projects from the handle **634**. The handle **634** helps support the eye bolt **632** and provides a grip to assist in installation or removal of a retainer element **620** on the fluid end body **602**.

(148) The cable **636** may be disposed through each eye **640** of each eye bolt **632**. Each of the eye bolts **632** may be oriented to facilitate the passage of the cable **636** through each eye **640**. The ends of the cable **636** may be attached to the external surface **604** of the fluid end body **602** using eye bolts **650** and clamps **652**. The cable **636** is preferably made of a stiff and tough material, such as high-strength nylon or steel.

(149) In operation, the eyebolts **632** and cable **636** tether each of the retaining elements **620** and components **614**, in case of failure of the retainer elements **620**, a portion of the fastening system **626**, or the fluid end body **602**. When a failure occurs, the large pressure in the fluid end body **602** will tend to force the components **614** out of their respective bores **608** with a large amount of energy. The cable **636** helps to retain the components **614** within the bores **608** in the event of a failure. The cable **636** also helps to retain the retainer elements **620** in position in the event of a failure. The fastening systems **134**, **222**, **320**, **420**, and **522** used with fluid ends **100**, **200**, **300**, **400**, and **500** may also be configured for use with the eye bolts **632**, handles **634** and cable **636**.

(150) In alternative embodiments, the handles **634** may not be used. A single eye bolt **632** may also be formed integral with a single component **614**. A single cable **636** may also be used through each of the eyebolts **632**. Each cable **636** would independently attach to the external surface **604** of the fluid end body **602**.

(151) Several kits are useful for assembling the fluid end **600**. A first kit comprises a plurality of the components **614**, a plurality of the retainer elements **620**, and the fastening system **626**. The kit may be assembled using the fluid end body **602**.

(152) Turning to FIGS. **16-20**, a seventh embodiment of the fluid end **800** is shown. The fluid end **800** comprises a housing or fluid end body **836**. The fluid end **800** is similar to the fluid ends **100** and **500**, shown in FIGS. **1-3**, **12** and **13**, with the below described exceptions.

(153) With reference to FIGS. **19**, **20**, and **30-32**, a component **804** installed within a suction bore **808** may also be referred to as a suction plug **804**. The suction bore **808** may also be characterized as a fourth section of the horizontal conduit. The suction plug **804** is sized to form a barrier to fluid flow within the suction bore **808**. Each of the suction plugs **804** comprises a cylindrical body having opposed top and bottom surfaces **816** and **818**. The suction plug **804** is of single-piece construction and is substantially solid with the exception of a threaded hole **820** formed in its top surface **816**. The suction plug **804** includes an upper or first portion **822** joined to a lower or second portion **824** by a tapered portion **827**.

(154) Continuing with FIGS. **30-32**, the upper portion **822** has an outer diameter, **D1** and the lower portion **824** has an outer diameter, **D2**. The lower portion **824** has a reduced diameter relative to that of the upper portion **822**, meaning the diameter **D1** is greater than the diameter **D2**. As shown in FIGS. **30-32**, the lower portion **824** has a length that is more than three times a length of the upper portion **822**. The lower portion **824** includes a plurality of sections along its length; the sections have several different diameters. The section of greatest diameter is situated midway along the length of the lower portion **824**, and presents an external sealing surface **826**. The sealing surface **826** may be considered the diameter **D2** of the lower portion **824**.

(155) First and second sections **828** and **830**, having outer diameters **D3** and **D4**, are formed on opposite sides of the sealing surface **826** and diameter **D2**. Each of the sections **828** and **830** has a reduced diameter relative to that of the sealing surface **826**. Thus, the diameter **D2** is greater than the diameters **D3** and **D4**. A third section **832** extends between the second section **830** and the bottom surface **818** and has an outer diameter, **D5**. The third section **832** has a reduced diameter relative to that of the second section **830**. Thus, the diameter **D4** is greater than the diameter **D5**.

(156) With reference to FIG. **21**, a plurality of beveled corners **834** are formed in the fluid end body **836** at the intersection of a front surface **838** and the walls surrounding the opening of each suction bore **808**. When a suction plug **804** is installed within one of the suction bores **808**, the tapered portion **827** of the plug **804** engages the beveled corners **834**. Such engagement prevents further axial movement of the plug **804** within the bore **808**. The upper portion **822** of the plug **804**, which includes diameter **D1**, projects from a front external surface **838** of the fluid end body **836** when installed within one of the bores **808**. Thus, upon installation of the suction plug **804** within the suction bore **808**, the difference between the diameters **D1** and **D2** prevents further axial movement of the suction plug **804** into the fluid end body **836** along the longitudinal axis of the suction bore **808**. In alternative embodiments, the upper portion of the suction plug may engage the front surface

of the fluid end body. In further alternative embodiments, axial movement of the suction plug within the bore may be prevented by engagement of the bottom surface of the plug with the walls surrounding the bore or a protrusion from the walls surrounding the bore.

(157) Turning back to FIGS. **30-32**, the outer surface of the plug **804** includes no annular recess for housing a seal. Instead, an annular recess **840** is formed in the walls surrounding each of the suction bores **808** adjacent the front surface **838** of the fluid end body **836**, as shown in FIG. **21**. The recess **840** is configured for housing an annular seal **842**. Preferably, the seal **842** is a high pressure seal.

(158) With reference to FIG. **25**, each recess **840** comprises two sidewalls **844** joined by a base **846**. The seal **842** is closely received within the recess **840**. After a seal **842** is installed within a corresponding recess **840** within a bore **808**, a suction plug **804** is installed within that bore.

(159) When a suction plug **804** is installed within a bore **808**, the seal **842** within the bore tightly engages the plug's sealing surface **826**. During operation, the seal **842** wears against the sealing surface **826** of the suction plug **804**. If the sealing surface **826** on one of the plugs **804** begins to erode, allowing fluid to leak around the plug **804**, that plug **804** is removed and replaced with a new plug. The seal **842** may also be removed and replaced with a new seal, if needed.

(160) Continuing with FIG. **25**, a small amount of clearance exists between the walls surrounding the bore **808** and the first, second, and third sections **828**, **830**, and **832** of the installed plug **804**. The clearance allows the suction plug **804** to rock back and forth on each side of its sealing surface **826**. The rocking motion helps to overcome friction between each of the plugs **804** and the walls surrounding its corresponding bore **808**. Thus, less force is required for installation or removal of one of the plugs **804** than is required for a traditional suction plug. Reduced forces mean fewer scrapes and scratches on the walls surrounding the bore, as compared to a traditional suction plug.

(161) The suction plugs **804** may be installed and removed using a tool (not shown), which may be attached to a plug **804** at the threaded hole **820**, shown in FIG. **30**. For example, a tool having an externally threaded end may mate with the internal threads formed in the threaded hole **820**. Once installed, an operator may rock the plug **804** back and forth using the tool while simultaneously pushing or pulling on the plug **804** with the tool.

(162) Continuing with FIGS. **19** and **20**, a component **806** installed within a discharge bore **810** may also be referred to as a discharge plug **806**. The discharge bore **810** may also be characterized as the first section of the vertical conduit. Turning to FIGS. **27-29**, each of the discharge plugs **806** comprises a cylindrical body having opposed top and bottom surfaces **850** and **852**. The discharge plug **806** is substantially solid with the exception of two threaded holes. A first threaded hole **854** formed in its top surface **850** and a second threaded hole **856** formed in its bottom surface **852**. Each plug **806** includes an upper portion **858** joined to a lower portion **860** by a tapered portion **862**. The lower portion **860** includes a plurality of sections along its length; the sections have several different diameters. The section of the greatest diameter is situated midway along the length of the lower portion **860**, and presents an external sealing surface **864**. First and second sections **866** and **868** are formed on opposite sides of the sealing surface **864**. Each of the sections **866** and **868** has a reduced diameter relative to that of the sealing surface **864**. A third section **870** is formed below the second section **868** and has a reduced diameter relative to that of the second section **870**. The third section **870** includes a plurality of reduced diameter sections.

(163) Each plug **806** further includes a connection portion **872**. The connection portion **872** extends between the third section **870** and the bottom surface **852**. The connection portion **872** has a reduced diameter relative to that of the lower portion **860**. The second threaded hole **856** extends within the connection portion **872**. The connection portion **872** is configured for connecting to, engaging, or aligning a spring **874** used with a discharge valve **876**, shown in FIGS. **19** and **20**.

(164) With reference to FIG. **22**, a plurality of beveled corners **878** are formed in the fluid end body **836** at the intersection of the top surface **880** and the walls surrounding the opening of each discharge bore **810**. When a discharge plug **806** is installed within one of the discharge bores **810**,

the tapered portion 862 of the plug 806 engages the beveled corners 878. Such engagement prevents further axial movement of the plug 806 within the bore 810. The upper portion 858 of the plug 806 projects from the top surface 880 of the fluid end body 836 when installed within one of the bores 810. In alternative embodiments, the upper portion of the discharge plug may engage the top surface of the fluid end body. In further alternative embodiments, axial movement of the discharge plug within the bore may be prevented by engagement of the bottom surface of the plug with the walls surrounding the bore or protrusions from the walls surrounding the bore.

(165) Turning back to FIGS. 27-29, the outer surface of the plug 806 includes no annular recess for housing a seal. Instead, an annular recess 882 is formed in the walls surrounding each of the discharge bores 810 adjacent the top surface 880 of the fluid end body 836, as shown in FIG. 22. The recess 882 is configured for housing an annular seal 884. Preferably, the seal 884 is a high pressure seal.

(166) With reference to FIG. 26, each recess 882 comprises two sidewalls 886 joined by a base 888. The seal 884 is closely received within the recess 882. After a seal 884 is installed within a corresponding recess 882 within a bore 810, a discharge plug 806 is installed within that bore.

(167) When a discharge plug 806 is installed within a bore 810, the seal 884 tightly engages the plug's sealing surface 864. During operation, the seal 884 wears against the sealing surface 864 of the discharge plug 806. If the sealing surface 864 on one of the plugs 806 begins to erode, allowing fluid to leak around the plug 806, that plug 806 is removed and replaced with a new plug. The seal 884 may also be removed and replaced with a new seal, if needed.

(168) Continuing with FIG. 26, a small amount of clearance exists between the walls surrounding the bore 810 and the first, second, and third sections 866, 868 and 870 of the installed plug 806. The clearance allows the discharge plug 806 to rock back and forth on each side of its sealing surface 864. The rocking motion helps to overcome friction between each of the plugs 806 and the walls surrounding its corresponding bore 810. The discharge plugs 806 may be installed and removed using a tool (not shown), which may be attached to a plug 806 at the threaded hole 854, shown in FIG. 27.

(169) With reference to FIGS. 19 and 20, when the fluid end 800 is operating, the bottom surfaces 818 and 852 of each of the plugs 804 and 806 will be exposed to the high fluid pressures within the interior of the fluid end 800. The fluid pressure may be high enough to dislodge the suction and discharge plugs 804 and 806 from their respective bores 808 and 810. To keep the plugs 804 and 806 within their respective bores 808 and 810, a plurality of retainers 890 are attached to the fluid end body 836. A retainer 890 is attached to the body 836 in front of each of the suction plugs 804 and above each of the discharge plugs 806, as shown in FIG. 20.

(170) With reference to FIG. 33, each retainer 890 has a cylindrical body having flat opposing top and bottom surfaces 892 and 894. A threaded central passage 896 is formed in the center of each retainer 890. The central passage 896 interconnects the top and bottom surfaces 892 and 894. A plurality of peripheral passages 898 are formed in each retainer 890 and surround the central passage 896. Each peripheral passage 898 interconnects the top and bottom surfaces 892 and 894 of each retainer 890.

(171) With reference to FIGS. 34 and 35, a retainer nut 900 is installed within the central passage 896 of each retainer 890, as shown in FIGS. 19 and 20. A central passage 902 is formed in the retainer nut 900. The central passage 902 interconnects the nut's top and bottom surfaces 904 and 906. External threads are formed on the retainer nut 900 adjacent its bottom surface 906. The external threads are matingly engageable with the internal threads formed in the retainer 890, as shown in FIGS. 19 and 20. The walls surrounding the central passage 902 adjacent the top surface 904 of the retainer nut 900 are shaped to closely receive a hex-shaped tool.

(172) With reference to FIG. 20, a plurality of peripheral openings 908 are formed in the fluid end body 836 around each opening of each suction and discharge bore 808 and 810. The peripheral passages 908 formed in each retainer 890 are alignable with the peripheral openings 908 formed

around each of the bores **808** and **810**, in a one-to-one relationship.

(173) Each of the retainers **890** is secured to the fluid end body **836** using a fastening system **910**, as shown in FIGS. **17** and **18**. The fastening system **910** comprises a plurality of studs **912**, a plurality of washers **914**, and a plurality of nuts **916**. Each stud **912** is externally threaded adjacent its first end **918**, while each peripheral opening **908** formed in the fluid end body **836** has internal threads that mate with those of the stud **912**, as shown in FIGS. **23** and **24**. Studs **912** are threaded into place within each of the peripheral openings **908**.

(174) Continuing with FIGS. **23** and **24**, once a first stud **912** has been installed in the fluid end body **836** at its first end **918**, its opposed second end **919** projects from the body's top or front surface **880** or **838**. Each peripheral passage **989** formed in each of the retainers **890** receives a corresponding one of the studs **912**. Each of the studs **912** receives a washer **914** and nut **916**, which hold the retainer **890** against the top and front surface **880** and **838** of the fluid end body **836**. Rather than applying a single large torque to a single retainer, the fastening system **910** contemplates distribution of smaller torques among a plurality of studs **912** and nuts **916**.

(175) When a retainer **890** is attached to the fluid end body **836**, the central passage **896** surrounds the upper portion **822** or **858** of the plug **804** or **806**. The retainer nut **900** installed within the retainer **890** is torqued so that its bottom surface **906** tightly engages with the top surface **816** or **850** of the plug **804** or **806**. Such engagement maintains the plug **804** or **806** within its corresponding bore **808** or **810**. When the retainer nut **900** is engaged with the top surface **816** or **850** of the plug **804** or **806**, the threaded hole **820** or **854** formed in the plug **804** or **806** is exposed to the nut's central passage **902**.

(176) During operation, an operator may need access to the inside of the fluid end **800** multiple times during a single fracking operation. For example, one of the plugs **804** or **806** may need to be replaced. Removing a retainer **890** to gain such access can be time-consuming, because of the need to remove multiple nuts **916** and washers **914**.

(177) To avoid such delays, each retainer **890** includes a removable retainer nut **900**. Rather than remove all of the nuts **916** and washers **914**, the operator can simply remove the retainer nut **900**. When the retainer nut **900** is removed, the operator can access the interior of the fluid end body **836** through the central opening **896** of the retainer **890**. The retainer nut **900** may be removed using a hex-shaped tool that mates with the walls surrounding the central passage **902** of the retainer nut **900**.

(178) While the fluid end **800** includes a plurality of threaded retainer nuts **900**, those retainer nuts **900** are not threaded into the walls surrounding the bores **808** and **810**. Thus, even if the threads on one of the retainer nuts **900** should crack, the fluid end body **836** remains intact. Only the retainer nut **900** and/or its corresponding retainer **890** need be replaced. The high cost of repairing or replacing the fluid end body **836** is thereby avoided.

(179) With reference to FIGS. **19**, **20**, **36** and **37**, a component **812** is installed within a plunger bore **814** of the fluid end **800**. The plunger bore **814** may be characterized as the third section of the horizontal conduit. The component **812** may be characterized as a stuffing box **812**. The stuffing box **812** has a cylindrical front portion **920** joined to a cylindrical rear portion **922**. The front portion **920** has a lesser diameter than that of the rear portion **922**. The stuffing box **812** is installed within the plunger bore **814** such that the front portion **920** is in front of or closer to the internal chamber **811** than the rear portion **922**.

(180) In contrast to the stuffing box sleeve **158** shown in FIG. **3**, the stuffing box **812** does not include a tapered portion **164**. Instead, the front portion **920** is joined directly to a front surface **924** of the rear portion **922**. The stuffing box **812** is also more robust than the sleeve **158**. However, one of skill in the art will recognize that the stuffing box **812** and the sleeve **158** serve the same purpose and are functionally interchangeable with one another.

(181) Continuing with FIGS. **36** and **37**, a central passage **926** extends through the stuffing box **812** and interconnects the box's rear and front surfaces **928** and **930**. An internal seat **932** is formed in

the walls surrounding the central passage **926** adjacent the front surface **924** of the rear portion **922**, as shown in FIG. **19**. The internal seat **932** is formed as a result of the rear portion **922** having a larger inner diameter from that of the front portion **920**.

(182) The rear portion **922** has a generally uniform outside diameter along its length. A plurality of peripheral passages **934** are formed in the rear portion **922** and surround the central passage **926**. The passages **934** interconnect the stuffing box's rear surface **928** and the front surface **924** of the rear portion **922**. A plurality of threaded openings **936** are formed in the rear surface **928** of the stuffing box **812**. The threaded openings **936** allow use of a tool for gripping the stuffing box **812** while it is being installed or removed.

(183) Turning back to FIG. **20**, when the stuffing box **812** is installed within the fluid end **800**, a lubrication hole **938** is formed in the fluid end body **836** and faces the stuffing box **812**. Because of the alignment between the lubrication hole **938** and the stuffing box **812**, first, second, and third annular recesses **940**, **942**, and **944** are formed in an outer surface of the stuffing box **812**, as shown in FIGS. **36** and **37**. Each of the first and third recesses **940** and **944** are configured to house a seal **946**, as shown in FIG. **19**. Preferably, the seal **946** is an O-ring. The second recess **942** underlies the lubrication hole **938**, and is interconnected with the stuffing box's central passage **926** by a plurality of spaced passages **948**. Lubrication may be supplied to the stuffing box **812** by way of the lubrication hole **938**, the second recess **942**, and the passages **948**.

(184) Turning back to FIGS. **36** and **37**, the outer surface of the stuffing box **812** includes no annular recess for housing a high pressure seal. Instead, an annular recess **950**, configured to house an annular seal **952**, is formed in the walls surrounding each plunger bore **814**, as shown in FIG. **19**. Preferably, the seal **952** is a high pressure seal.

(185) Continuing with FIG. **19**, each recess **950** is characterized by two side walls **954** joined by a base **956**. The width of the base **956** is equal to at least one-third of the length of the front portion **920** of the stuffing box **812**. The recess **950** is sized so as to house a large seal **952**, much larger than an O-ring. The seal **952** is closely received within the recess **950** and may fill at least 95% of the volume of the recess **950**. After a seal **952** is installed within a recess **950** formed within one of the plunger bores **814**, a stuffing box **812** is installed within that bore **814**.

(186) When a stuffing box **812** is installed within a bore **814**, the seal **952** tightly engages the outer surface of the stuffing box's front portion **920**. During operation, the seal **952** wears against the front portion **920**. If the outer surface of the front portion **920** begins to erode, allowing fluid to leak around the stuffing box **812**, that stuffing box **812** can be removed and replaced with a new stuffing box. The seal **952** may also be removed and replaced with a new seal, if needed.

(187) The seals **167** and **250** shown in FIGS. **3** and **6** may be identical to the seal **952**. Likewise the seal **167** and **250** may be positioned in a recess that is identical to the recess **950**.

(188) Continuing with FIG. **20**, a counterbore **958** is formed within the plunger bore **814**. A plurality of openings **960** are formed in a base **962** of the counterbore **958**. When a stuffing box **812** is installed within the plunger bore **814**, the front surface **924** of the rear portion **922** engages the base **962** of the counterbore **958**. Such engagement prevents further movement of the stuffing box **812** within the fluid end body **836**. The stuffing box **812** is positioned within the plunger bore **814** such that its peripheral passages **934** and the openings **960** formed in the base **962** are aligned in a one-to-one relationship, as shown in FIG. **20**.

(189) With reference to FIGS. **38** and **39**, a retainer **970** prevents the stuffing box **812** from being dislodged from the plunger bore **814**. The retainer **970** comprises a cylindrical body having an internally threaded central passage **972**. The central passage **972** interconnects the retainer's rear and front surfaces **974** and **976**. A plurality of peripheral passages **978** surround the central passage **972** and interconnect the retainer's rear and front surfaces **974** and **976**. A counterbore **980** is formed within each passage **978**, adjacent the top surface **974** of the retainer **970**.

(190) With reference to FIG. **20**, the retainer **970** is installed within the plunger bore **814** so that its front surface **976** engages the rear surface **928** of the stuffing box **812**. The retainer **970** is installed

over the stuffing box **812** such that the peripheral passages **934** and the peripheral passages **978** are aligned in a one-to-one relationship.

(191) Each of the retainers **970** is secured to the fluid end body **836** using a fastening system **982** shown in FIGS. **17** and **18**. The fastening system **982** comprises a plurality of studs **984** and a plurality of nuts **986**. Each of the studs **984** is received within a corresponding one of the openings **960** formed in the base **962**. From the base **962**, each stud **984** extends through a corresponding one of the passages **934** in the stuffing box **812**, and through a corresponding one of the passages **978** in the retainer **970**.

(192) A first end **988** of each stud **984** is positioned within one of the counterbores **980** formed in the retainer **970**. A nut **986** is then placed on a second end **987** of each stud **984**, and turned until it tightly engages the base of the counterbore **980**. In alternative embodiments, the fastening system may comprise a plurality of screws instead of studs and nuts. The screws are preferably socket-headed cap screws. In further alternative embodiments, the fastening system may comprise a plurality of bolts, such as 12-point bolts.

(193) Attaching the retainer **970** to the fluid end body **836** also helps ensure the stuffing box **812** remains tightly in place during operation. Because each of the retainers **970** is attached to the fluid end body **836** using the fastening system **982**, no external threads are formed on the outer surface of each of the retainer **970**. Likewise, no internal threads are formed within the walls of each plunger bore **814**.

(194) Continuing with FIGS. **19** and **20**, a plunger packing **990** is installed within the central passage **926** of each stuffing box **812**. When installed, the plunger packing **990** engages the stuffing box's internal seat **932**. Thus, the plunger packing **990** is only positioned within the rear portion **922** of the stuffing box **812** and is spaced away from the front portion **920** and the seal **952**. The plunger packing **990** prevents high pressure fluid from passing around a plunger **991** as the plunger reciprocates. Each plunger packing **990** comprises a plurality of annular seals compressed together and having aligned central passages. The outer seals **994** may be made of metal and compress the inner pressure seals **996**, as shown in FIG. **19**. The inner pressure seals **996** are preferably high pressure seals. The seal **952** is approximately the same width as one of the pressure seals **996**.

(195) Over time, the seals **994** and **996** wear against the inner surface of the stuffing box **812**. If leakage occurs, the stuffing box **812** may be removed and replaced with a new stuffing box.

(196) The plunger packing **990** is held within the stuffing box **812** by a packing nut **992**. The packing nut **992** is generally identical to the packing nut **182** shown in FIGS. **2** and **3**. However, the packing nut **992** may vary slightly in size from the packing nut **182** in order to properly fit within the retainer **970** and stuffing box **812**. External threads formed on the outer surface of the packing nut **992** matingly engage the internal threads formed in the retainer **970**. An O-ring may be installed within a groove **993** formed within the packing nut **992**, as shown in FIG. **19**.

(197) When a packing nut **992** is installed within one of the retainers **970**, a front surface **998** of the packing nut **992** engages the plunger packing **990**. Such engagement compresses the plunger packing **990**, creating a tight seal. After a packing nut **992** has been installed within a retainer **970**, a central passage within that packing nut **992** will be aligned with a central passage in a plunger packing **990**.

(198) When assembling the fluid end **800**, the stuffing box **812** is installed within the plunger bore **814** after installation of the seal **952** within the recess **950**. After the stuffing box **812** is installed, the retainer **970** may be secured to the housing using the fastening system **982**. After the retainer **970** is attached to the fluid end body **836**, the plunger packing **990** may be installed within the rear portion **922** of the stuffing box **812**. The plunger packing **990** is installed through the rear surface **928** of the stuffing box **812**. Alternatively, the plunger packing **990** may be installed within the stuffing box **812** prior to attaching the retainer **970** to the fluid end body **836**. During operation, the plunger packing **990** may be removed and replaced with a new plunger packing **990** without removing the stuffing box **812** from the fluid end **800**.

(199) Once a stuffing box **812**, plunger packing **990**, retainer **970**, and packing nut **992** are installed within a plunger bore **814**, a plunger **991** is next installed, as shown in FIG. **20**. Alternatively, the plunger **991** may be installed prior to installing the packing nut **992**. Once installed, the plunger **991** is surrounded by the other components within the plunger bores **814**. During operation, the plunger **991** moves relative to the fluid end **800** and the components installed within the plunger bores **814**.

(200) The plunger **991** is identical to the plunger **188** shown in FIGS. **2** and **3**. A clamp **1005** is attached to the end of each plunger **991**. The clamp **1005** secures its plunger **991** to a pony rod **1007**, shown in FIG. **40**.

(201) Continuing with FIG. **40**, the fluid end **800** is shown attached to a power end **1009** via a plurality of stay rods **1011**. The power end **1009** drives reciprocal motion of the plungers **991** within the fluid end **800**. The fluid ends **100**, **200**, **300**, **400**, **500**, or **600** may be attached to the power end **1009** in place of the fluid end **800**.

(202) Several kits are useful for assembling the fluid end **800**. A first kit comprises a plurality of the components **804**, a plurality of the components **806**, a plurality of the components **812**, a plurality of the retainers **890**, a plurality of the retainers **970**, and the fastening systems **910** and **982**. The kit may be assembled using the fluid end body **836**.

(203) With reference to FIGS. **1-14** and **16-20**, a single fluid end body may use any combination of the kits described herein. The fluid end bodies, components, and retainer elements described herein are preferably made of high strength steel.

(204) While the fluid end bodies **102**, **202**, **302**, **402**, **502** and **836** shown in FIGS. **1-13** and **16-20** are substantially rectangular in shape, the kits described herein may also be used with any shape of a fluid end body, such as that shown in FIG. **14**. Likewise, the retainer elements described herein may vary in shape and size, as desired. For example, the circular retainer elements described herein may be square or rectangular shaped.

(205) The fastening systems **134**, **222**, **320**, **420**, **522** and **910** described herein each use eight studs around each bore opening. In alternative embodiments, more than eight studs or less than eight studs may be used to secure each retainer element over each bore opening. For example, FIG. **14** only shows six studs securing each retainer element **620** over each bore opening **610**. Likewise, fewer than 16 or more than 16 screws may be used with the fastening systems **178**, **260**, **352**, **444**, and **982**. The number of peripheral openings formed in each retainer element described herein may correspond with the number of openings formed around each bore opening in each fluid end body and the number of studs or screws being used.

(206) The fastening systems described herein reduce the amount of torque required to secure each retainer element to the fluid end bodies. Rather than having to torque one large retaining nut, the torque is distributed throughout the plurality of studs, nuts, or screws. Decreasing the amount of torque required to seal the bores increases the safety of the assembly process.

(207) Turning to FIG. **15**, a stud **700** is shown. The stud **700** may be used with the fastening systems **134**, **222**, **320**, **420**, **522**, **626**, and **910** shown in FIGS. **1**, **4**, **7**, **9**, **11**, **14**, **17** and **18**. For exemplary purposes, the stud **700** will be described with reference to fluid end **100**, shown in FIG. **1**.

(208) The stud **700** has a first threaded section **702** and an opposite second threaded section **704**. The threaded sections **702** and **704** are joined by an elongate, cylindrical body **706**. The first threaded section **702** is configured for threading into one of the plurality of threaded openings **144** formed in the fluid end body **102**. The second threaded section **704** is configured for threading into the threaded opening formed in one of the nuts **152**.

(209) The first section **702** may have fewer threads than that of the opening **144**. For example, if the opening **144** has 18 internal threads, the first section **702** of the stud **700** may only have 16 external threads. This configuration ensures that all of the threads formed on the first section **702** will be engaged and loaded when the first end **702** is threaded into the opening **144**. Engaging all of

the threads helps to increase the fatigue life of the first end **702** of the stud **700**. Likewise, the second section **704** may have fewer external threads than there are internal threads formed in the nut **152**. The stud **700** may also be subjected to shot peening on its non-threaded sections prior to its use to help reduce the possibility of fatigue cracks. The stud **700** may have a smooth outer surface prior to performing shot peening operations.

(210) The body **706** of the stud **700** comprises a first section **708** and a second section **710**. The first section **708** has a smaller diameter than the second section **710**. The retainer element **132** is primarily held on the first section **708** of the body **706**. The diameter of the second section **710** is enlarged so that it may center the washer **150** on the stud **700**.

(211) The diameter of the second section **710** is configured so that it is only slightly smaller than the diameter of the central opening of the washer **150**. This sizing allows the washer **150** to closely receive the second section **710** of the stud **700** when the washer **150** is positioned on the stud **700**. When the washer **150** is positioned on the second section **710**, the washer **150** is effectively centered on the stud **700**. The washer **150** is also effectively centered against the nut **152**, once the nut **152** is installed on the stud **700**.

(212) Without placing the washer **150** on the second section **710**, the washer may have to be manually centered on the stud **700** prior to installing the nut **152**. If the washer **150** is not properly centered on the stud **700** or against the nut **152**, it may be difficult to effectively torque or un-torque the nut **152** from the stud **700**, depending on the type of washer used.

(213) The plurality of washers used with each fastening system **134**, **222**, **320**, **420**, **522**, **626** and **910** shown in FIGS. **1**, **4**, **7**, **9**, **11**, **14**, **17** and **18** may be configured to allow a large amount of torque to be imposed on the nuts used with the washers without using a reaction arm. Instead, the washer itself may serve as the counterforce needed to torque a nut onto a stud. Not having to use a reaction arm increases the safety of the assembly process. The nuts used with the fastening systems **134**, **222**, **320**, **420**, **522**, and **626** may also comprise a hardened inner layer to help reduce galling between the threads of the nuts and studs during the assembly process. An example of the above described washers, nuts, and methods are described in Patent Cooperation Treaty Application Serial No. PCT/US2017/020548, authored by Junkers, et al, the entirety of which is incorporated herein by reference.

(214) With reference to FIGS. **41-54**, an eighth embodiment of a fluid end **1000** is shown. The fluid end **1000** comprises a housing or fluid end body **1002**. The fluid end **1000** is identical to the fluid end **800**, but comprises an alternative embodiment of a stuffing box **1004** and a corresponding retainer **1006**. The stuffing box **1004** and retainer **1006** are each installed within a plunger bore **1008**. The plunger bore **1008** is identical to the plunger bore **814** formed in the fluid end **800** and shown in FIGS. **19** and **20**.

(215) With reference to FIGS. **44-47**, the stuffing box **1004** is similar to the stuffing box **812** shown in FIGS. **36** and **37**. The stuffing box **1004** comprises a first portion **1010** joined to a second portion **1012**. The first and second portions **1010** and **1012** each have a cylindrical shape, such that the stuffing box **1004** may be considered primarily cylindrical. The first portion **1010** has an outer diameter, **D1**. The second portion **1012** has an outer diameter, **D2**. The diameter **D2** is greater than the diameter **D1**. The diameter **D2** is also greater than a maximum diameter of a groove **1014** formed in the housing **1002**, as shown in FIGS. **41-43**. Like the stuffing box **812**, the stuffing box **1004** is installed within the plunger bore **1008** such that the first portion **1010** is closely received within a first segment **1016**, and the second portion **1012** is closely received within a second segment **1018** of the plunger bore **1008**, as shown in FIG. **41**. The difference between the diameters **D1** and **D2** and the diameters of the plunger bore **1008** prevent further movement of the stuffing box **1004** into the housing **1002**, as shown in FIG. **41**.

(216) Continuing with FIGS. **44-47**, the diameter **D1** is constant along at least a portion of the length of the first portion **1010** of the stuffing box **1004**. The diameter **D1** may be constant along the entire length of the first portion **1010**, with the exception of a tapered surface **1020** formed

between the first portion **1010** and a first surface **1022** of the stuffing box **1004**. No grooves are formed in the outer surface of the first portion **1010** for housing a seal. Rather, the outer surface of the first portion **1010** has a sealing surface for a seal **1024**, as shown in FIGS. **41-43**. The diameter **D2** is constant along the entire length of the second portion **1012**. In alternative embodiments, the diameter **D2** may only be constant along a portion of the length of the second portion **1012**.

(217) Continuing with FIGS. **44-47**, the first portion **1010** has an inner diameter, **D3**. The second portion **1012** has an inner diameter, **D4**. The diameter **D4** is greater than the diameter **D3**. The diameter **D3** may be constant along at least a portion of the length of the first portion **1010**, and the diameter **D4** may be constant along the length of the second portion **1012**.

(218) Continuing with FIG. **47**, the inner surface of the first portion **1010** may have a slightly convex portion **1026** joined to a straight portion **1028**. The convex portion **1026** may extend between an internal seat **1030** and the straight portion **1028**. Because the first portion **1010** includes the convex portion **1026**, the first portion **1010** may also have an inner diameter, **D5**. The diameter **D3** is greater than the diameter **D5**. The convex portion **1026** helps increase the wall thickness of the first portion **1010**, which helps alleviate stress within the stuffing box **1004** during operation. In alternative embodiments, the inner surface of the first portion **1010** may be shaped like the stuffing box **812** shown in FIGS. **19** and **20**.

(219) With reference to FIGS. **44-46**, like the stuffing box **812**, a plurality of passages **1032** are formed in the second portion **1012** and interconnect a first and a second surface **1034** and **1036** of the second portion **1012**. The passages **1032** are configured for receiving a plurality of fasteners **1038**, as shown in FIG. **42**. The fasteners **1038** may be threaded studs or socket-headed screws and are part of a fastening system, like the fastening system **982** shown in FIGS. **17** and **18**.

(220) In contrast to the stuffing box **812**, a plurality of annular recesses are not formed in an outer surface of the second portion **1012**. Instead, as will be discussed in more detail below, such recesses are formed in the retainer **1006**. Because the recesses are formed in the retainer **1006**, the second portion **1012** of the stuffing box **1004** has a shorter length than that of the stuffing box **812**, shown in FIGS. **36** and **37**. Likewise, the retainer **1006** has a longer length than the retainer **970**, shown in FIGS. **38** and **39**.

(221) With reference to FIGS. **48-54**, the retainer **1006** comprises a cylindrical body having a central passage **1040**. The central passage **1040** interconnects the retainer's first and second surfaces **1042** and **1044**. A plurality of peripheral passages **1046** surround the central passage **1040** and interconnect the retainer's first and second surfaces **1042** and **1044**. The passages **1046** are configured for receiving the fasteners **1038**, as shown in FIG. **42**. The retainer **1006** is attached to the housing **1002** using a fastening system in the same manner as the retainer **970**. The retainer **1006** is attached to the housing **1002** such that the first surface **1042** of the retainer **1006** abuts the second surface **1036** of the stuffing box **1004**.

(222) Continuing with FIGS. **48-54**, a first, second, and third annular recess **1050**, **1052**, and **1054** are formed in an outer surface of the retainer **1006**. Each of the first and third recesses **1050** and **1054** are configured to house a seal (not shown). Preferably, the seal is an O-ring. A plurality of passages **1056** interconnect the central passage **1040** and the second recess **1052**. Each of the passages **1056** extends at a non-zero angle relative to a longitudinal axis extending through the central passage **1040**. The opening of each of the passages **1056** in the second recess **1052** is positioned closer to the second surface **1044** of the retainer **1006** than the opening of the passage **1056** within the central passage **1040**. In alternative embodiments, the passages **1056** may be formed at a different angle from that shown in FIGS. **41** and **51**. A portion of the central passage **1040** adjacent the second surface **1044** of the retainer **1006** is threaded and configured to engage a packing nut **1060**, as shown in FIG. **41**.

(223) Turning back to FIG. **41**, when the stuffing box **1004** is installed within the housing **1002**, the second recess **1052** underlies a lubrication conduit **1062** formed in the housing **1002**. The lubrication conduit **1062** interconnects an external surface **1003** of the housing **1002** and the

plunger bore **1008**. A plurality of lubrication conduits **1062** may be formed in the housing **1002**, each conduit corresponding with one of the plunger bores **1008**.

(224) In operation, lubricant is pumped into the housing **1002** through the conduit **1062**. The lubricant enters the second recess **1052** and flows circumferentially around the retainer **1006** until it encounters one of the passages **1056**. Lubricant flows through one or more of the passages **1056** and into a plunger packing **1066** installed within the retainer **1006** and the stuffing box **1004**. The lubricant lubricates a plunger **1068** as it reciprocates. The angled nature of the passages **1056** help the flow of lubricant from the housing **1002** to the plunger packing **1066**.

(225) With reference to FIGS. **43**, **46**, **47**, and **54**, a plurality of blind bores **1070** are formed in the second surface **1036** of the stuffing box **1004**. Each bore **1070** is configured for receiving a dowel pin **1072**, as shown in FIG. **43**. A plurality of corresponding bores **1074** are formed in the first surface **1042** of the retainer **1006** for also receiving the dowel pins **1072**. The dowel pins **1072** keep the retainer **1006** aligned with the stuffing box **1004** during assembly, and help carry any non-tensile loading imparted by the interface between the retainer **1006** and the stuffing box **1004**.

(226) The bores **1074** formed in the retainer **1006** and shown in FIGS. **43** and **54** interconnect the first and second surfaces **1042** and **1044** of the retainer **1006**. In alternative embodiments, the bores **1074** may be blind bores that open on the first surface **1042** of the retainer **1006**. The bores **1070** and **1074** may include counter-bored areas configured for receiving the dowel pins **1072**, as shown in FIGS. **43**, **47**, and **54**.

(227) The various features and alternative details of construction of the apparatuses described herein for the practice of the present technology will readily occur to the skilled artisan in view of the foregoing discussion, and it is to be understood that even though numerous characteristics and advantages of various embodiments of the present technology have been set forth in the foregoing description, together with details of the structure and function of various embodiments of the technology, this detailed description is illustrative only, and changes may be made in detail, especially in matters of structure and arrangements of parts within the principles of the present technology to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. An apparatus, comprising: a plunger; and a fluid end assembly, comprising: a housing having an external surface with a plurality of first openings and a plurality of opposed plunger openings formed therein; a plurality of bore pairs formed within the housing, each bore pair comprising: a straight first bore and a straight second bore, the second bore intersecting the first bore and interconnecting one of the first openings and one of the plunger openings; a valve pair situated within a first bore of a selected bore pair; a stuffing box sleeve installed within a second bore of the selected bore pair; and a first plate fastened to the external surface of the housing and covering the plurality of first openings; and a second plate fastened to the external surface of the housing, the second plate having a plurality of plate bores axially aligned with the plurality of plunger openings; in which the stuffing box sleeve is retained within the second bore by the second plate; and in which the second bore of the selected bore pair at least partially receives the plunger.
2. The apparatus of claim 1, further comprising: a plurality of fasteners; in which the first plate comprises a plurality of openings; in which each of the plurality of fasteners extends through one of the plurality of openings and into the housing to secure the first plate to the housing.
3. The apparatus of claim 1, further comprising a suction plug situated within the second bore of the selected bore pair; in which a top surface of the suction plug sits flush with the external surface of the housing.
4. The apparatus of claim 3, in which the suction plug is a separate piece than the first plate.
5. The apparatus of claim 1, further comprising a discharge plug situated within the first bore of the

selected bore pair.

6. The apparatus of claim 1, in which the external surface of the housing includes at least one flat surface.

7. The apparatus of claim 1, in which the external surface of the housing comprises a plurality of threaded openings formed in groups, each group positioned around one of the first openings.

8. The fluid end assembly of claim 7, further comprising a plurality of fasteners, the plurality of fasteners passing through the first plate and into the plurality of threaded openings in a one-to-one relationship.

9. The fluid end assembly of claim 1, in which no threads are formed on the stuffing box sleeve.

10. A fluid end assembly, comprising: a housing, comprising: an external surface, comprising: a first face having a plurality of first openings formed therein; and a second face having a plurality of second openings formed therein; a plurality of first and second bores formed therein, each first bore intersecting a corresponding second bore, each second bore joining a first opening and a second opening; a valve pair situated within one of the first bores; a stuffing box sleeve situated within one of the second bores; and a retaining plate fastened to the second face of the housing; and a second plate fastened to the external surface of the housing, the second plate having a plurality of plate bores axially aligned with the plurality of first openings; in which the stuffing box sleeve is retained within the second bore by the second plate; and in which each first opening is configured to receive at least a portion of a plunger.

11. The fluid end assembly of claim 10, further comprising a seal positioned around the stuffing box sleeve, the seal configured to block fluid from leaking from the selected second bore.

12. The fluid end assembly of claim 10, further comprising a plurality of fasteners; in which the retaining plate comprises a plurality of passages; in which each of the plurality of fasteners extends through one of the plurality of passages and into the housing to secure the retaining plate to the housing; in which each of the plurality of fasteners is a threaded fastener; and in which each of the plurality of fasteners includes a threaded nut.

13. The fluid end assembly of claim 10, in which no internal threads are formed within the second openings.

14. The fluid end assembly of claim 10, in which no threads are formed on the stuffing box sleeve.

15. The fluid end assembly of claim 10, in which the housing is made of two sections joined together by a plurality of fasteners; in which each of the two sections receives at least a portion of the plunger.

16. The fluid end assembly of claim 10, in which the first and second faces are spaced and parallel.

17. An apparatus, comprising: a housing, comprising: an external surface that includes a plunger surface and a flat remote surface; a first bore extending between different points on the external surface; a plurality of second bores extending between the plunger surface and the remote surface and comprising: a plurality of second bore openings formed in the plunger surface; and a plurality of second bore openings formed in the remote surface; in which a selected one of the second bores intersects the first bore; a pair of valves installed within the first bore on opposite sides of the first bore's intersection with the selected second bore; a plunger at least partially received within the selected second bore and extending through one of the second bore openings in the plunger surface; and a flat plate disposed in overlaying engagement with the remote surface such that each second bore opening in the remote surface is covered; and a second plate fastened to the plunger surface of the housing, the second plate having a plurality of plate bores axially aligned with the plurality of second bore openings; in which the plunger extends through the second plate.

18. The apparatus of claim 17, in which the selected one of the second bores is straight.

19. The apparatus of claim 17, in which the plunger surface and the remote surfaces are spaced and parallel.

20. The apparatus of claim 17, further comprising a suction plug situated in the selected second bore.

21. The apparatus of claim 20, in which the suction plug does not extend past the remote surface of the external surface.
